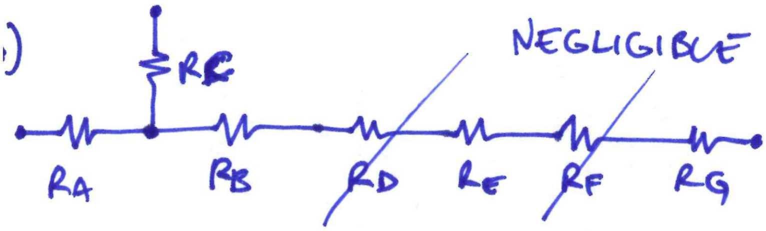
EX. 1

1)



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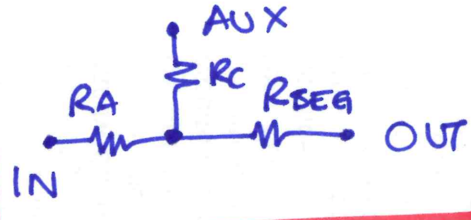


$h = 20$ $\quad W = 100$ $\qquad \gamma = \dfrac{12 \cdot \eta}{\left(1 - 0.63\frac{1}{5}\right) \cdot h^3 \cdot W} = 1.71 \cdot 10^{16}$

20, 20 $\qquad r_{eq} = \dfrac{2 \cdot e^2}{4e} = \dfrac{e}{2} \longrightarrow \beta = \dfrac{8 \cdot \eta}{\pi r^4} = 2.54 \cdot 10^{17}$

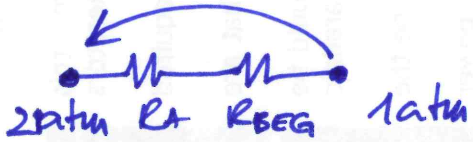
$R_A = \gamma \cdot A = \boxed{3.43 \cdot 10^{13}}$ Pa·s/m³

$R_{BEG} = \gamma \cdot (\overset{1}{B} + \overset{2}{E} + \overset{1}{G}) = 2 \cdot R_A = \boxed{6.86 \cdot 10^{13}}$ Pa·s/m³

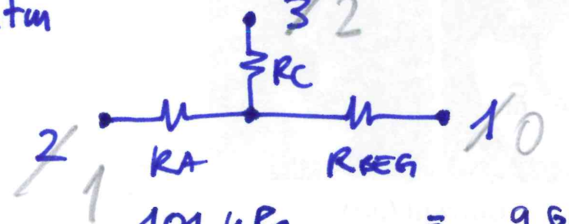
$R_C = \beta \cdot C = \boxed{5 \cdot 10^{14}}$

when VALVE CLOSED ⟶ speed = ∅

VALVE OPEN — AUX CLOSED



VALVE OPEN — AUX = 3atm



2) $v = \dfrac{Q}{h \cdot w}$ $\qquad Q = \dfrac{\Delta P}{R_A + R_{BEG}} = \dfrac{101\ kPa}{(3.43 + 6.86) \cdot 10^{13}} = 9.81 \cdot 10^{-10}\ \dfrac{m^3}{s}$

$v = \dfrac{9.81 \cdot 10^{-10}}{20\mu m \cdot 100\mu m} = \boxed{0.49\ m/s}$

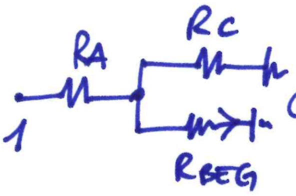
3) SUPERPOSITION:

$Q = Q_1 + Q_2 =$

$= 11.13 \cdot 10^{-10}\ m^3/s$

$\boxed{v = 0.55\ m/s}$

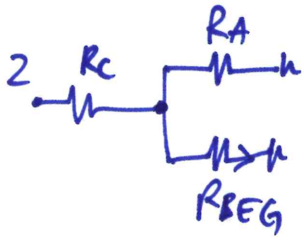
Q_1



$\dfrac{6.86 \cdot 50}{6.86 + 50} = 6$

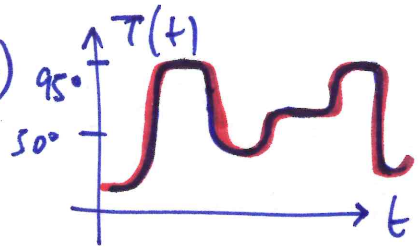
$Q_1 = \dfrac{1\ atm \cdot R_{//}}{R_A + R_{//}} \cdot \dfrac{1}{R_{BEG}} = 9.36 \cdot 10^{-10}$

Q_2



$\dfrac{3.43 \cdot 50}{3.43 + 50} = 3.2$

$Q_2 = \dfrac{2\ atm \cdot R_{//}}{R_{//} + R_C} \cdot \dfrac{1}{R_{BEG}} = 1.77 \cdot 10^{-10}$

b) 

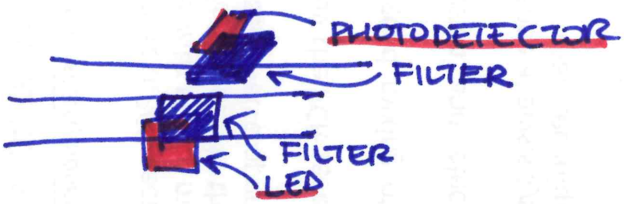
PCR

RANGE $10 \div 95\,°C$ ACCURACY $1\,°C$

c) MIXERS $<$ ACTIVE / PASSIVE

d) $D = \frac{kT}{6\pi\eta r}$ Diff. coeff $\propto T$ ⇧

$\mu = \frac{q}{6\pi r \eta}$ $\eta = \eta(T)$ ⇧

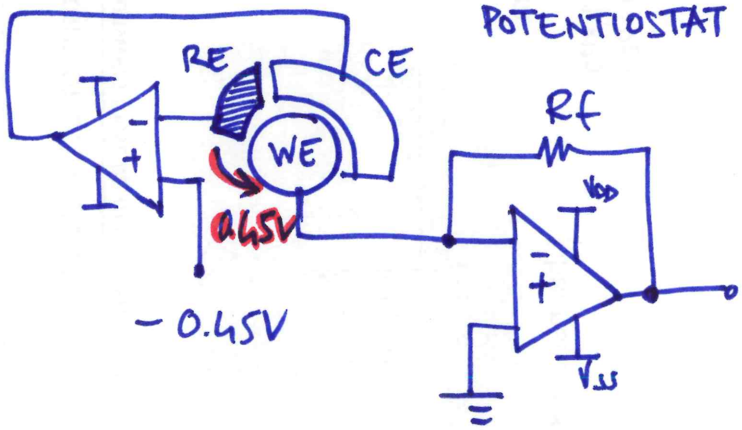
e) 

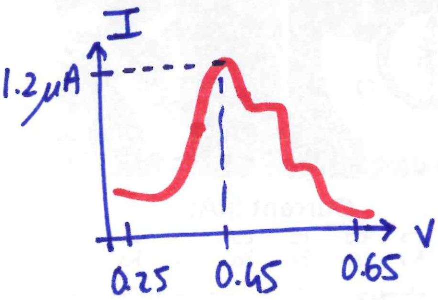
) CELL LYSIS < MECHANICAL / THERMAL / CHEMICAL

PRO	CON
	HIGH PRESSURE
NO POWER	CONTAMINATION

EX. 2

POTENTIOSTAT

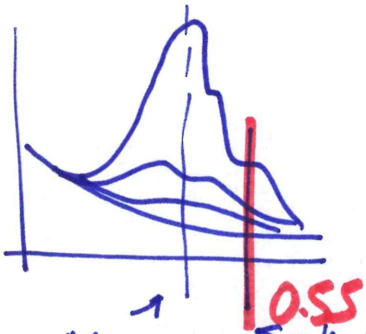




$R_f = \frac{5V}{1.2\mu A} = 4.16\ M\Omega \longrightarrow \boxed{4M\Omega}$ some margin

b) $BW = 40 \cdot SR = 40 \cdot 100\frac{V}{s} = 4kHz$

check $\frac{1}{2\pi\, 4M\Omega \cdot 0.2pF} = 198\ kHz$ ok c)

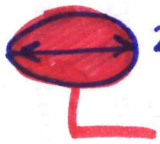


I choose 0.55 V

MAX ↗ ← More signal @ LOW CONCENTRATIONS

LIMIT: INTERFERENCE

d)



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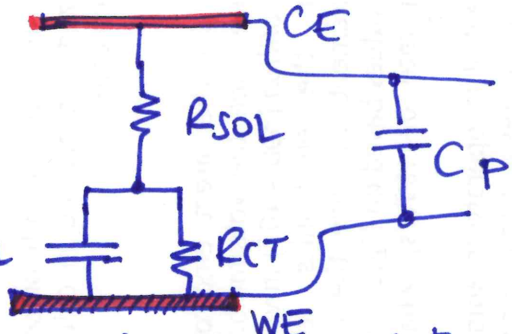


can quantify $\begin{cases} C_{DL} \text{ from AREA} & \frac{0.1\,pF}{\mu m^2} \cdot \pi \cdot (1000\,\mu m)^2 = \boxed{314\ nF} \\ R_{CT} \text{ from VOLTAMMOGRAM} & \left.\frac{\partial I}{\partial V}\right|_{\bar{V}}^{-1} = \end{cases}$

$\bar{V} = 0.3 \Rightarrow \left(\frac{0.4\,\mu A}{0.05\,V}\right)^{-1}$ $\quad R_{CT} = \frac{50\,mV}{0.4\,\mu A} = \boxed{125\ k\Omega}$

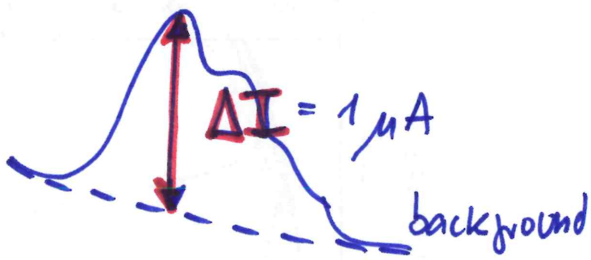
e)

$\sigma_i = \sqrt{\frac{4kT}{R} \cdot BW}$ $\begin{cases} BW = 4\,kHz \text{ from point (b)} \\ BW = \frac{1}{2\pi R \cdot 0.2\,pF} = 1.6\,MHz \cdot \frac{\pi}{2} = 2.5\,MHz\ B_{eq} \end{cases}$

(TOO LARGE, USELESS)

$\sigma_i \begin{cases} 11.5\ pA_{RMS} \\ 288.4\ pA_{RMS} \end{cases}$

from plot



$LoD = \frac{11.5\,pA}{1\,\mu A} \cdot 100\,ppb = \boxed{0.0011\ ppb}$

) $V_{DD} \downarrow \Rightarrow R_f \downarrow \Rightarrow$ NOISE UP ↑

) INSULATING SUBSTRATE (SiO_2)

THICK PASSIVATION ⟶ REDUCE C_{stray}